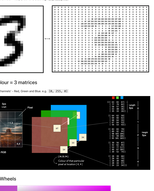
09 Data Libraries - Numpy, Pandas, Seaborn

Plan for this lecture

1. Matrices and Computer Vision
1. Pandas
2. Reading Data files (CSVs) - Pandas
3. Data Visualisation (Seaborn / Matplotlib)
4. AI examples - iTunes API

0.0 The importance of matrices for Computer Vision!

- Images are 2D arrays of numbers!
- Grayscale = one "channel" of 0 to 255 values.
- 0 = White
- 255 = Black
- RGB = three "channels" - Red, Green and Blue, e.g. [R, [G], B]



0.1 RGB Colour = 3 matrices

- RGB = Red Intensity! - Red, Green and Blue, e.g. [R, [G], B]

Red
Green
Blue
Pixel
Pixel
Pixel

```
[0.0, 0.0, 1.0, 0.0, 'Iris-setosa'],
```

mean	7.5850000	4.400000	3.500000
std	0.8280661	0.435866	2.565459

In [5]:
```
iris_df['petal_width'].mean()
```

Out[5]:
```
1.1993333333333333
```

In [6]:
```
iris_df['petal_width'].std()
```

Out[6]:
```
0.7622376689603465
```

In [7]:
```
iris_df['petal_width'].max()
```

Out[7]:
```
2.5
```

In [8]:
```
iris_df['petal_width'].min()
```

Out[8]:
```
0.1
```

In [9]:
```
iris_df['petal_width'].mean()
```

Out[9]:
```
1.1993333333333333
```

Let's use Seaborn for data visualisation

In [10]:
```
import seaborn
```

```
python3 -m pip install --U seaborn --user
```

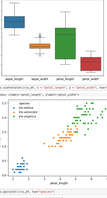
In [11]:
```
import seaborn as sns
```

Out[11]:
```
[]
```

In [12]:
```
module("seaborn", from="/Users/miniforge3/envs/Python3.9/lib/python3.9/site-packages/seaborn/__init__.py")
```

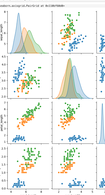
In [13]:
```
sns.boxplot(iris_df)
```

Out[13]:
```
<Axes: >
```



In [14]:
```
sns.scatterplot(iris_df, x="petal_length", y="petal_width", hue="species")
```

Out[14]:
```
<Axes: <class'pandas.core.frame.DataFrame'>,'petal_length','petal_width',hue='species'>
```



In [15]:
```
sns.pairplot(iris_df, hue="species")
```

Out[15]:
```
<seaborn.axisgrid.PairGrid at 0x7f8c8f8d8d0>
```

4.0 APIs and Data Science - iTunes example

Example searches:

- https://itunes.apple.com/search?term=jack+johnson
- https://itunes.apple.com/search?term=jack+johnson&limit=25
- https://itunes.apple.com/search?term=jack+johnson&limit=25&offset=25

Not just music: iTunes has books, which you can lookup via ISBNs:

- https://itunes.apple.com/lookup?isbn=9780596008678

For more information and examples of how to construct search queries, see the iTunes documentation here.

Depending on which environment you're running this notebook, you may need to install `requests`, before using the package.

```
pip install requests
```

```
python3 -m pip install --U requests
```

In [16]:
```
import requests
```

In [17]:
```
def search_itunes(query, media,limit=None):
    '''requests itunes and return results in JSON format.

    Arguments:
        query (str): The search term to query for (e.g. music, movies, podcasts).
        media_type (str): The type of media to search for (e.g. music, movies, podcasts).

    Keyword Arg:
        limit: The JSON response from the iTunes API.

    Usage:
        >>> r = search("jack+johnson", "music")
        >>> len(r.keys())
        4
        >>> len(r['results'])
        25 # iTunes defaults to the number of results to 25

    '''
    response = requests.get(base_url, params=params)
    # print(response.status_code)
    # print(response.content)
    # print(response.text)
    return response.json() # Convert the response to JSON

if __name__ == "__main__":
    # Example usage
    r = search("jack+johnson", "music")
    keys = r.keys()
    results = len(r['results'])
    print(keys)
    print(results)
```

Out[17]:
```
{'resultCount': 50,
 'results': [{'artistId': 1141707,
   'artistName': 'Jack Johnson',
   'collectionCensoredName': 'Sing-Along Songs for Healthy Living Vol. 1 (Clean Version)',
   'collectionExplicitness': 'notExplicit',
   'collectionId': 1440857781,
   'collectionName': 'Sing-Along Songs for Healthy Living Vol. 1',
   'collectionPrice': 9.99,
   'collectionViewUrl': 'https://music.apple.com/us/album/sing-along-songs-for-healthy-living-vol-1/1440857781?uo=4',
   'contentAdvisoryRating': 'Clean',
   'country': 'USA',
   'currency': 'USD',
   'primaryGenre': {'name': 'Children's Music',
    'nameForms': {'plural': 'Children's Music'},
    'subgenreId': '4'},
   'trackCount': 10,
   'trackNamesFormsShortVersion': {'plural': 'Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs, Healthy Living Songs'},
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  {'artistId': 1141707,
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```

```
In [ ]: # Create your array here.
```

Exercise 2

Create a `numpy.ndarray` of seven zeros.

```
In [ ]: # Create your array here.
```

Exercise 4

Create a `numpy.ndarray` with values 0 - 10.

```
In [ ]: # Create your array here.
```

Exercise 5

Take a look at `numpy.arange` with only with numbers between range of 50 and 100.

```
In [ ]: # Write your solution here.
```

Exercise 6:

Take a look at `reshape`. Use the following one-dimensional array (`ndr`) into a two-dimensional array (5 x 5).

```
In [1]: ndr = np.arange(25)
        ndr
Out[1]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14,
               15, 16, 17, 18, 19, 20, 21, 22, 23, 24])
```

```
In [2]: # Write your solution here.
```

```
Out[2]: array([[ 0,  1,  2,  3,  4],
               [ 5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14],
               [15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24]])
```

Exercise 7

Off the reshaped two-dimensional array, how would you print just the middle column?

```
In [3]: # Write your solution here.
Out[3]: array([ 2,  7, 12, 17, 22])
```

Exercise 10

Create a one-dimensional `numpy.ndarray`. It should be three rows by three columns (3 x 3). The values should be between the integer values 0 and 9, allowed randomly.

Run the cell a few times to see if the values in your 3X3 are generated randomly each time.

```
In [ ]: # Write your solution here.
```

Exercise 9

Starting with the same two-dimensional array - see if you can write some exception handling code to respond to `IndexError`.

```
In [4]: ndr = np.array([[0,1,2],[3,4,5],[6,7,8]])
        ndr
Out[4]: array([[0, 1, 2],
               [3, 4, 5],
               [6, 7, 8]])
```

Write some exception handling code to respond appropriately to index values below the lower-bound, and values above the upper-bound. For example below:

```
In [5]: ndr[3,0] # Below the upper bound of the array
```

```
Traceback (most recent call last)
Indexerror: index 3 is out of bounds for axis 0 with size 3
```

```
In [6]: ndr[-1,0] # Below the lower bound of the array
```

```
Traceback (most recent call last)
Indexerror: index -1 is out of bounds for axis 0 with size 3
```

Exercise 10

In preparation for our game, write code to print the `X` symbols arranged in a pyramid like below:

```
In [ ]: # Write your solution here.
```

```
X
XX
XXX
XXXX
XXXXX
```

```
In [ ]: # Extensions within a 20 array!
```

```
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
```

Exercise 11

Use your file handling knowledge to write the data contained in objects of a class (e.g., `Student`) to write each student name and id in a separate line (row) in a text file (e.g., `students.txt`).

Start by completing the Student class below:

```
In [ ]: class Student:
            def __init__(self, name, id):
                self.name = name # creates this line
                self.id = id # creates this line
                # and so on with attributes
```

Now see if you can create three objects of this class. Replace the `...` below to complete the statements:

```
In [ ]: s1 = Student(...) # create your first student object here
        s2 = Student(...) # create your second student object here
        s3 = Student(...) # create your third student object here
```

Now see if you can write the student attributes out to a file. Complete the below:

```
In [ ]: file_pointer = open("students.txt", "w") # for write - create the file

        file_pointer.write(s1.name + "\n") # write the attributes data for your first student object here.
        file_pointer.write(s1.id + "\n") # write the attributes data for your first student object here.
        file_pointer.write(s2.name + "\n") # write the attributes data for your second student object here.
        file_pointer.write(s2.id + "\n") # write the attributes data for your second student object here.
```

Extension: Which exceptions / error handling would you add to the above code?

Exercise 12

Now see if you can read the data back into a Pandas DataFrame, which should format the comma-separated values as cells in a table. You should see each student's data on one, and the name, id and mark as columns.

First check that you have access to `pandas`, by running the cell below:

```
In [8]: import pandas as pd
```

```
In [9]: !ls -la students.txt
```

```
-rw-r--r-- 1 jamesmf staff 1 Jan 8 18:24 students.txt
```

Providing you have pandas installed, see if you can use a particular method in `read` to read from your `students.txt`, and store in a `pandas.DataFrame`.

Note, four `IndexerrorS` errors, or worse, will be raised when attempting the statements in `...`, but this particular `read` method should work for both CSV files and txt files.

```
In [10]: students_df = ... # read in your students.txt file here
         # print the name of the dataframe here to print it
         students_df
```

```
Out[10]:  name  id  mark
0  Sam  SC0001  70  NaN
1  Jan  SC0002  72  NaN
2  Jan  C00000  74  NaN
```

Extension: Add the data of the more student objects to the file (`StudentTwo`. Now calculate the maximum mark of these students - which of your students achieved the mark? Can you identify the two?

```
In [ ]: # Write your solution here.
```

Exercise 13

Read in the dataset as a numpy array with the code below. Can you extract the three line flower labels from the 'species' column ('iris setosa', 'iris versicolor', and the virginica) and code these back in a dictionary, which maps to `0`, `1`, and `2` respectively, representing each class?

```
In [11]: !wget -q -O iris.csv https://raw.githubusercontent.com/mwitcher/datasets/master/iris.csv
         import numpy as np, sys, os, urllib.request, urllib.error, urllib.parse
         names = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']
         data = np.genfromtxt('iris.csv', delimiter=',', dtype=None, names=names)
         print(data)
```

```
[(5.1, 3.5, 1.4, 0.2, 'Iris-setosa')
 (4.9, 3.0, 1.4, 0.2, 'Iris-setosa')
 (4.7, 3.2, 1.3, 0.2, 'Iris-setosa')
 (4.6, 3.1, 1.5, 0.2, 'Iris-setosa')
 (5.0, 3.6, 1.4, 0.2, 'Iris-setosa')
 (5.4, 3.9, 1.7, 0.4, 'Iris-setosa')
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 (5.0, 3.4, 1.5, 0.2, 'Iris-setosa')
 (4.4, 2.9, 1.4, 0.2, 'Iris-setosa')
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 (5.8, 4.0, 1.2, 0.2, 'Iris-setosa')
 (5.7, 4.4, 1.5, 0.4, 'Iris-setosa')
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```

See below the attempt to format the data returned from the `search_itunes()` function to format the data into a DataFrame.

Question: What do you notice about the results column below?

In [3…

```
#from search_itunes import search_itunes #only if you have a .py script for search_itunes

df = pd.DataFrame(search_itunes("Taylor", "music"))
df
```

Out[3…

	resultCount	results
0	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
1	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
2	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
3	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
4	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
5	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
6	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
7	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
8	10	{'wrapperType': 'track', 'kind': 'song', 'arti...
9	10	{'wrapperType': 'track', 'kind': 'song', 'arti...

Your task is to transform the above DataFrame into the format shown below:

In [3…

```
# Write your solution here...
```

Out[3…

	wrapperType	kind	artistId	collectionId	trackId	artistName	collectionName	trackName	collectionCensoredName	trackCensoredName	…	di
0	track	song	400555769	1506831755	1506832357	Natalie Taylor	Surrender - Single	Surrender	Surrender - Single	Surrender	…	
1	track	song	400555769	1182561824	1182561980	Natalie Taylor	In the Air Tonight - Single	In the Air Tonight	In the Air Tonight - Single	In the Air Tonight	…	
2	track	song	400555769	880114237	880114240	Natalie Taylor	Love Life - Single	Love Life	Love Life - Single	Love Life	…	
3	track	song	400555769	1519882762	1519882765	Natalie Taylor	Surrender (Kina Remix) - Single	Surrender (Kina Remix)	Surrender (Kina Remix) - Single	Surrender (Kina Remix)	…	
4	track	song	400555769	1178290096	1178290111	Natalie Taylor	Loved You Before - Single	Loved You Before	Loved You Before - Single	Loved You Before	…	
5	track	song	645420096	1572859144	1572859302	ILLENIUM & Natalie Taylor	Fallen Embers	Fragments	Fallen Embers	Fragments	…	
6	track	song	400555769	978021284	978021290	Natalie Taylor	Wildfire - EP	Wildfire	Wildfire - EP	Wildfire	…	
7	track	song	400555769	1457378386	1457378387	Natalie Taylor	Iris - Single	Iris	Iris - Single	Iris	…	
8	track	song	400555769	1154603331	1154603480	Natalie Taylor	Amen - Single	Amen	Amen - Single	Amen	…	
9	track	song	400555769	988419300	988419846	Natalie Taylor	I Want It All - Single	I Want It All	I Want It All - Single	I Want It All	…	

10 rows × 32 columns

Extension: Write code which returns the song with the longest `trackTime`

Hint: Start by identifying the column that has data on the `trackTime`

In […

```
```

Exercise 16

Find an CSV file online, that you can import via `read_csv()`. Be aware that the data needs to be in raw format (without the html/css/table structure). Alternatively, download the CSV file and read it locally (as some sites won't give you the raw data in a readable/scrapable format).

Examples of CSV data:

https://github.com/riteshc6/amazon_scraper/blob/master/downloads/Computers%20%26%20Accessories.csv

https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv

In [1…

```
url = "https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv"

df = pd.read_csv(url)

df
```

Inspect the data from your chosen CSV, and perform some exploratory data analysis - use Seaborn and/or Matplotlib to visualise trends.
Question: What trends do you see?

This website does not host notebooks. It only renders notebooks available on other websites.
Delivered by Fastly, Rendered by OVHcloud
nbviewer GitHub repository.
nbconvert version: 7.16.6
Rendered a minute ago